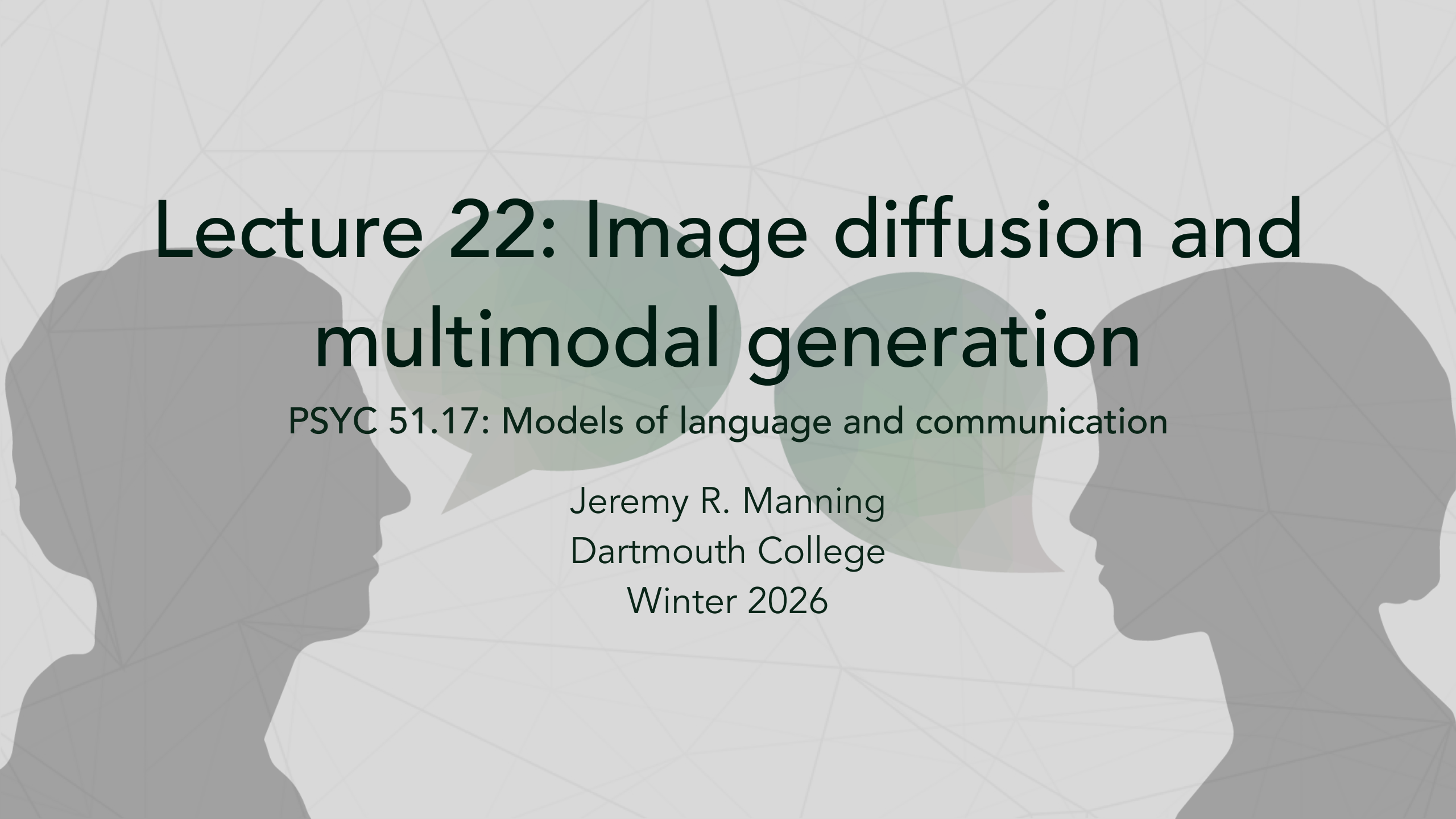




Lecture 22: Image diffusion and multimodal generation

PSYC 51.17: Models of language and communication

Jeremy R. Manning
Dartmouth College
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Learning objectives

By the end of this lecture, you will be able to...

1. Explain how **latent diffusion** compresses computation via a VAE to enable high-resolution image generation
2. Describe how **CLIP** creates a shared embedding space for text and images
3. Explain how **cross-attention** and **classifier-free guidance** enable text-controlled image generation
4. Compare the **U-Net** and **Transformer (DiT)** backbones for diffusion models
5. Generate images from text prompts using the HuggingFace `diffusers` library

From text to images

Bridging Lecture 21 and today

In Lecture 21, we saw how diffusion models generate **text** by iteratively unmasking tokens (discrete diffusion). Today we turn to the domain where diffusion was born — **images** — and ask: how do we generate high-resolution images from text prompts?

The key challenges

Challenge	Solution (this lecture)
Images are huge (786K+ values)	Latent diffusion — compress first, diffuse in latent space
Text must control generation	CLIP + cross-attention — bind words to spatial regions
Output must match the prompt	Classifier-free guidance — amplify text alignment
Architecture must scale	U-Net or DiT — backbones for the denoising network

The pixel problem

Why pixel-space diffusion is expensive

The DDPM framework (Lecture 21) can be applied to images by adding Gaussian noise to pixels and learning to reverse the process. But for a 512×512 RGB image, the denoising network must process $512 \times 512 \times 3 = 786,432$ values at every step — far too expensive for high-resolution generation.

U-Net (brief preview)

The **U-Net** ([Ronneberger et al., 2015](#)) is an encoder-decoder neural network with skip connections, originally designed for image segmentation. It became the default backbone for diffusion models — we'll cover it in detail later this lecture.

The numbers

Resolution	Pixels	Memory	Time per image
64×64	12,288	~2 GB	~30 seconds
256×256	196,608	~8 GB	~5 minutes
512×512	786,432	~32 GB	~20 minutes

The solution: don't run diffusion in pixel space. Run it in a **compressed latent space**.

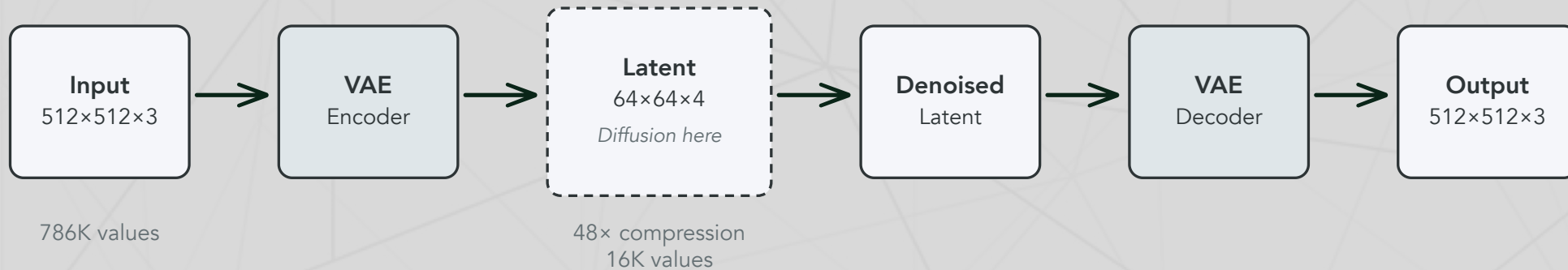
Latent diffusion

Compressing before diffusing

Latent diffusion separates image generation into two stages:

1. **Compression:** A pretrained **variational autoencoder** (VAE) encodes images into a compact latent space (typically $8\times$ spatial compression)
2. **Generation:** The diffusion process operates entirely in this latent space

The VAE is described on the next slide.



Further reading

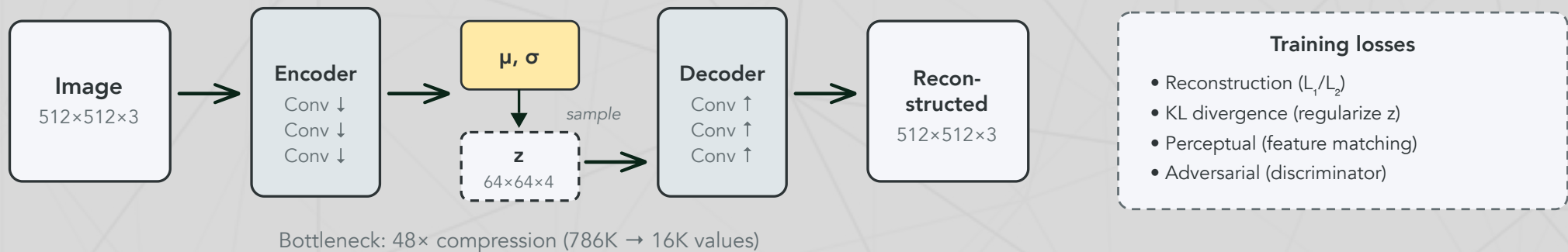
Rombach et al. (2022, CVPR). "High-resolution image synthesis with latent diffusion models" — The paper that introduced latent diffusion and enabled Stable Diffusion.

Variational autoencoder (VAE)

Compressing images to a latent space

A **VAE** learns to compress images into a low-dimensional latent representation and reconstruct them:

1. **Encoder:** Maps image $\mathbf{x} \in \mathbb{R}^{512 \times 512 \times 3}$ to mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\sigma}^2$ of a latent distribution
2. **Sampling:** Draw $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$ — a latent vector in $\mathbb{R}^{64 \times 64 \times 4}$
3. **Decoder:** Reconstructs the image from $\mathbf{z}$



Why this works

The VAE learns to preserve perceptually important information while discarding redundant detail. Diffusion in latent space is **~50× cheaper** than in pixel space, with negligible quality loss. This single insight enabled Stable Diffusion — the first open-source, consumer-GPU image generator.

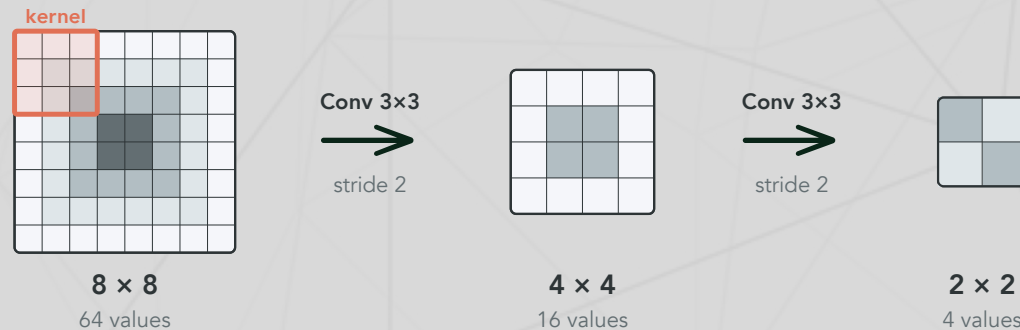
How convolutions compress images

Convolution: the key building block

A **convolution** slides a small filter (kernel) across an image, computing a weighted sum at each position. With **stride 2**, the filter moves 2 pixels at a time, halving the spatial dimensions:

- **8×8** input → 3×3 conv, stride 2 → **4×4** output (4× fewer values)
- **4×4** → 3×3 conv, stride 2 → **2×2** (4× fewer again)

Stacking convolution layers creates a hierarchy: pixels → edges → textures → objects.



What happens at each step

8×8: each cell = one pixel of the image

4×4: each cell summarizes a 2×2 region

2×2: each cell summarizes a 4×4 region

Each layer's receptive field grows — deeper layers "see" more of the original image.

The intuition

Each cell in a deeper layer "sees" a larger region of the original image — its **receptive field** grows. A 2×2 feature map from an 8×8 input means each cell summarizes a 4×4 patch. This is exactly how the VAE encoder compresses: stacked convolutions progressively trade spatial detail for compact, meaningful features.

The VAE bottleneck

Stable Diffusion's VAE in numbers

Property	Value
Input resolution	$512 \times 512 \times 3$
Latent resolution	$64 \times 64 \times 4$
Compression ratio	$48\times$ (786K $\rightarrow$ 16K values)
Reconstruction quality	Near-lossless for natural images

The VAE is trained once and frozen. The diffusion model only ever sees latents — it never touches pixels during training or sampling.

How the VAE is trained

- **Reconstruction loss** (L_1/L_2): Ensure decoded image matches the original
- **KL divergence**: Regularize the latent space so it's smooth and continuous
- **Perceptual loss**: Compare *features* extracted by a pretrained network (e.g., VGG), not raw pixels — so the model prioritizes visual similarity over pixel-exact matching
- **Adversarial loss**: A discriminator network tries to distinguish real from reconstructed images, pushing the decoder toward photorealistic outputs

CLIP: connecting text and images

Contrastive language-image pre-training

CLIP learns a **shared embedding space** for text and images. It was trained on 400 million image-text pairs from the internet:

1. **Image encoder** (Vision Transformer): Maps an image to a vector
2. **Text encoder** (Transformer): Maps a caption to a vector
3. **Contrastive training**: Matching pairs are pulled close; non-matching pairs are pushed apart

Why CLIP matters for diffusion

CLIP provides the "language understanding" for text-to-image systems. When you type "a cat wearing a hat," CLIP's text encoder converts it into an embedding that captures the **visual meaning** — not just the linguistic meaning. This embedding then guides diffusion via cross-attention.

Further reading

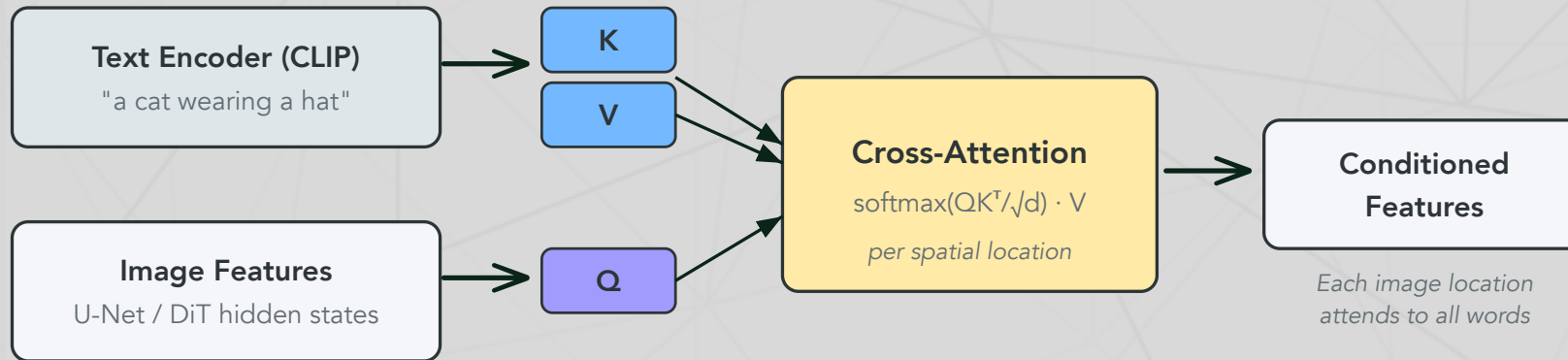
Radford et al. (2021, *ICML*) "Learning transferable visual models from natural language supervision" — Connection to Lecture 11: CLIP extends the idea of **learned embeddings** (Word2Vec, GloVe) to a *joint* text-image space.

Text conditioning with cross-attention

How text controls image generation

To generate images from text, the denoising backbone adds **cross-attention** layers (recall attention from Lecture 15). At each layer:

1. The text prompt is encoded by CLIP into a sequence of token embeddings
2. The image features at that layer serve as **queries** (Q)
3. The text embeddings serve as **keys** (K) and **values** (V)
4. Cross-attention allows each spatial location in the image to attend to relevant words



How 'a cat wearing a hat' becomes an image

The word "cat" activates high attention weights in the spatial region where the cat is generated. The word "hat" activates weights near the top of the cat region. This spatial-linguistic binding is learned entirely from image-caption pairs.

Classifier-free guidance

Controlling generation quality

Classifier-free guidance (CFG) improves alignment between text and generated images. During training, the text condition is randomly dropped some fraction of the time. At inference:

$$\tilde{\epsilon} = \underbrace{\epsilon_{\theta}(\mathbf{x}_t, t, \emptyset)}_{\text{unconditional}} + w \cdot \left(\underbrace{\epsilon_{\theta}(\mathbf{x}_t, t, c)}_{\text{conditional}} - \underbrace{\epsilon_{\theta}(\mathbf{x}_t, t, \emptyset)}_{\text{unconditional}} \right)$$

where w is the **guidance scale** (typically 7–15).

The intuition

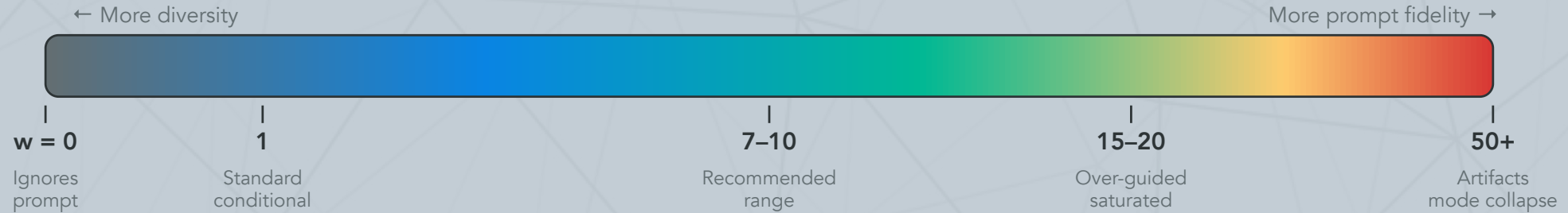
Think of CFG as asking: "What's *different* about images that match this prompt versus random images?" Then amplifying that difference. The model learns both what a "dog on a beach" looks like *and* what a random image looks like — guidance amplifies the gap. See the next slide for examples.

Further reading

Ho & Salimans (2022). "Classifier-free diffusion guidance" — Used in virtually all modern text-to-image systems.

The guidance scale tradeoff

What different guidance scales produce



In practice

Nearly every text-to-image system uses CFG. Stable Diffusion defaults to $w = 7.5$; DALL-E 2 uses $w \approx 4$. The range of 7–10 balances prompt fidelity against visual quality — pushing higher creates oversaturated, artifact-prone images.

Try it: varying the guidance scale

```
1 from diffusers import StableDiffusionPipeline
2 import torch
3
4 pipe = StableDiffusionPipeline.from_pretrained(
5     "runwayml/stable-diffusion-v1-5", torch_dtype=torch.float16
6 ).to("cuda")
7
8 for scale in [1.0, 7.5, 20.0]:
```

continued...

The guidance scale tradeoff

Try it: varying the guidance scale

```
9 img = pipe("a cat in a hat", guidance_scale=scale).images[0]  
10 img.save(f"guidance_{scale}.png")
```

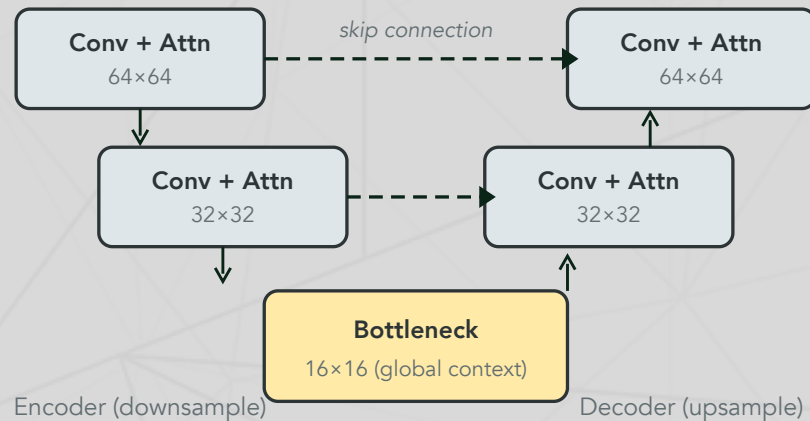
...continued

U-Net: the original diffusion backbone

An encoder-decoder with skip connections

The U-Net is the default backbone for diffusion models from DDPM through Stable Diffusion 1–2. Its U-shaped design provides **multi-scale reasoning**:

1. **Encoder** (downsampling): Reduces spatial resolution while increasing channels — captures global context
2. **Bottleneck**: Lowest resolution, largest receptive field
3. **Decoder** (upsampling): Restores spatial resolution — generates fine details
4. **Skip connections**: Connect encoder to decoder at matching resolutions — preserve high-frequency spatial information



Conditioning inputs

- Timestep t (sinusoidal embedding)

Text conditioning

- CLIP embeddings via cross-attention

Key properties

- Same-size input/output
- Multi-scale reasoning
- Skip connections preserve high-frequency detail
- ~860M params (SD 1.5)

Further reading

Ronneberger et al. (2015, *MICCAI*) "U-Net: convolutional networks for biomedical image segmentation" — Originally for medical imaging, now the workhorse of diffusion.

U-Net in diffusion models

How the U-Net is adapted for diffusion

Property	Configuration
Input/output	Same-sized latent (e.g., $64 \times 64 \times 4$)
Timestep conditioning	Sinusoidal embeddings added to each block
Text conditioning	CLIP embeddings injected via cross-attention layers
Skip connections	Concatenate encoder features to decoder
Parameters	~860M (Stable Diffusion 1.5)

The intuition: why multi-scale matters

The U-Net's encoder-decoder structure lets it reason at **multiple scales simultaneously**. The bottleneck captures high-level composition ("a dog sits on the left, a tree on the right") while skip connections preserve fine details ("the dog's fur texture, the tree's leaves"). This is why U-Nets produce images with both coherent structure *and* crisp details.

Connection to Lecture 15

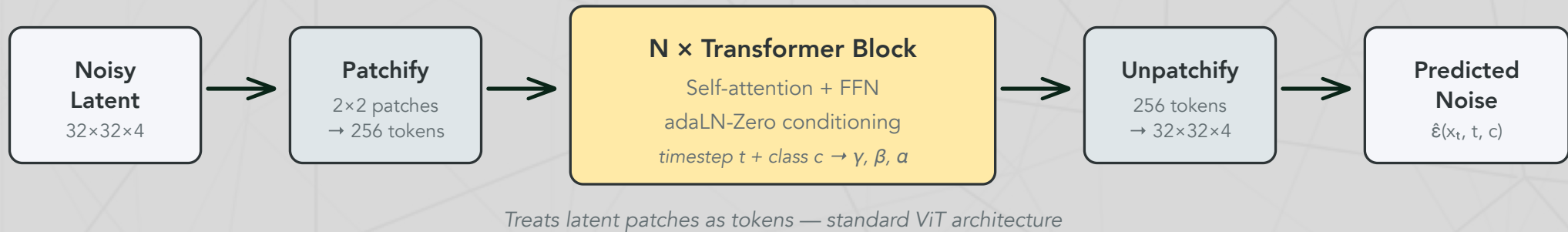
Cross-attention in the U-Net works exactly like encoder-decoder attention in the original Transformer (Lecture 15) — the image features are the "decoder" queries, and the text embeddings are the "encoder" keys and values.

Diffusion Transformer (DiT)

Replacing U-Net with a Transformer

The Diffusion Transformer (DiT) replaces the U-Net with a standard Vision Transformer:

1. **Patchify**: Divide the noisy latent into non-overlapping patches (e.g., 2×2)
2. **Flatten**: Treat patches as a sequence of tokens (just like ViT treats image patches — Lecture 15)
3. **Process**: Apply standard Transformer blocks with self-attention
4. **Unpatchify**: Reshape back to spatial dimensions to get the predicted noise



Key terms

- **Noisy latent**: The VAE-compressed image with Gaussian noise added at timestep t — this is the input the denoising network must "clean up"
- **Predicted noise**: The network's estimate of what noise was added — subtract it from the noisy latent to get a cleaner image

DiT: why replace U-Net?

Transformers scale better

DiT-XL/2 (675M parameters) achieves state-of-the-art FID of 2.27 on ImageNet, beating all previous diffusion models. More importantly, DiT shows **clean scaling behavior** — larger models consistently produce better results, with no architectural bottlenecks.

The intuition

U-Nets have strong *inductive biases* for spatial data (locality, hierarchy). These help with small models but become constraints at scale. Transformers make fewer assumptions and let the model learn the right structure from data — the same lesson we saw with language models (Lectures 15–16).

Further reading

Peebles & Xie (2023, ICCV), "Scalable diffusion models with Transformers" — DiT showed that Transformers can replace U-Nets in diffusion, and the result scales better.

Try it: generate an image

Text-to-image with Stable Diffusion (Google Colab)

```
1  !pip install -q diffusers transformers accelerate
2
3  import torch
4  from diffusers import StableDiffusionPipeline
5
6  pipe = StableDiffusionPipeline.from_pretrained(
7      "runwayml/stable-diffusion-v1-5", torch_dtype=torch.float16
8  ).to("cuda")
9
10 image = pipe("a photo of an astronaut riding a horse on mars").images[0]
11 image.save("astronaut.png")
```

What's happening under the hood

1. CLIP encodes your text prompt into embeddings
2. The U-Net iteratively denoises a random latent, guided by those embeddings via cross-attention
3. CFG amplifies the text signal at each step (default `guidance_scale=7.5`)
4. The VAE decoder converts the final latent back to a 512×512 pixel image

Every concept from this lecture is working together in those 5 lines of code.

Take-home messages

Key ideas from today

- The key bottleneck in high-resolution generation wasn't the diffusion process itself — it was **where** you run it. Compressing to latent space via a VAE made consumer-GPU generation possible.
- Text-to-image generation requires a **bridge between modalities**: CLIP creates a shared embedding space (extending the idea of word embeddings from Lecture 11), and cross-attention lets the diffusion model "listen" to text at every spatial location (using the same mechanism as encoder-decoder attention from Lecture 15).
- Classifier-free guidance shows that **controlling** generation is as important as generation itself — the trick is surprisingly simple: learn what conditional and unconditional outputs look like, then amplify the difference.
- The field evolves by **composing** innovations (VAE + CLIP + CFG + U-Net/DiT), not replacing them. Each addresses one specific limitation.

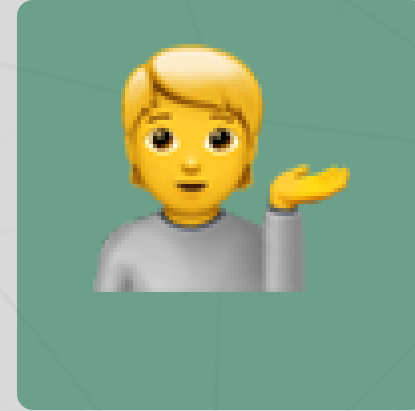
Questions?



Email



Discord



Office hours

Up next...

Diffusion applications: text-to-image (DALL-E 2, Imagen), text-to-video (Sora), and the ethics of generative AI